

Interim Data Analysis:

Associations Between Clinical Predictors and Orbital Cellulitis Admissions

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Introduction

Orbital cellulitis (OC) is a serious infection of the orbital tissues that can lead to vision-threatening and life-threatening complications, often requiring hospitalization and, in severe cases, surgical intervention. Clinical presentations can vary, ranging from localized inflammation to the development of orbital and intracranial complications such as subperiosteal abscess (SPA), orbital abscess (OA), and intracranial extension (ICE).

The data for these analyses consist of hospital admissions for OC from June 2006 through May 2025, derived from a retrospective chart review across three hospital systems: Denver Health (DH), University of Colorado (UC), and Oregon Health and Science University (OHSU). The dataset includes 330 adult subjects with detailed clinical, demographic, and outcome information collected at the time of admission.

The primary objectives of this analysis were to identify factors associated with need for surgical intervention and length of hospital stay (LOS) among adults admitted for orbital cellulitis. Secondary analyses examined factors associated with the development of complications, including SPA, OA, and ICE. We hypothesized that markers of disease severity and underlying comorbidities would be associated with increased likelihood of surgery and prolonged LOS.

Methods

2.1 Data Cleaning

Each row of the dataset represented a single observation of orbital cellulitis admissions.

Age was recalculated using date of admission and date of birth. One subject with sex recorded as “Other” was set to missing to improve model stability. Sex was then coded as a binary factor of Female and Male.

Race and ethnicity indicators were collapsed into a single race/ethnicity variable to reduce sparse categories. Hispanic ethnicity was prioritized regardless of race. Among non-Hispanic subjects, race/ethnicity was categorized as White non-Hispanic, Black non-Hispanic, or Other. The Other category included Asian/Pacific Islander, Native American, other race, and individuals who could not be classified into the White or Black non-Hispanic groups. We acknowledge that race is a social construct and that observed differences likely reflect structural inequities rather than biological differences.

Duplicate surgery-related variables were renamed to distinguish prior surgery from surgery during hospital admission. Final categorical predictors were coded as binary factors with “No” as the reference category and “Yes” as the comparison category.

2.2 Data Analysis

Descriptive statistics were used to summarize baseline characteristics. Continuous variables were summarized using means and standard deviations or medians and interquartile ranges, as appropriate, and categorical variables were summarized using counts and proportions.

To evaluate factors associated with need for surgical intervention, multivariable Firth penalized logistic regression models were used. This approach was selected to address potential issues of separation given the presence of sparse predictors and imbalanced outcome categories. Adjusted odds ratios (ORs) with 95% confidence intervals (CIs) and p-values were reported.

Factors associated with the development of complications (SPA, OA, and ICE) were also assessed using separate multivariable Firth logistic regression models for each outcome.

To evaluate factors associated with length of stay (LOS), multivariable negative binomial regression was used due to the count nature of LOS and evidence of overdispersion. Results were reported as rate ratios (RRs) with 95% confidence intervals. Rate ratios were interpreted as multiplicative and percentage changes in expected LOS.

Candidate predictors included demographic variables (age, sex, race/ethnicity) and clinical factors (sinusitis, diabetes, immunocompromised status, substance use, asthma, trauma, prior surgery, dental infection, nasolacrimal duct obstruction, SPA, OA, and ICE). Variables were selected a priori based on discussion with primary investigator (PI).

Sensitivity analyses were conducted by comparing models with and without inclusion of complication variables to assess potential mediation effects

All analyses used complete-case data under a missing at random assumption, where missingness depends on observed variables. Statistical significance was assessed at a two-sided $\alpha = 0.05$ level. All analyses were conducted in R (version 4.5.3; RStudio, Boston, MA).

2.2.1 Correlation

Prior to model fitting, pairwise associations among predictor variables were evaluated to assess potential collinearity. Pearson correlation coefficients were used for exploratory assessment of associations among variables. For binary variables, this is equivalent to the phi coefficient. A correlation matrix and heatmap visualization were generated to summarize relationships among predictors. Correlations greater than 0.7 were considered indicative of substantial shared variability.

Results

A total of 330 adult patients were included in the analysis. Baseline demographic and clinical characteristics are presented in Supplementary Tables 1–4 in the Supplementary section of this report.

3.1 Correlations

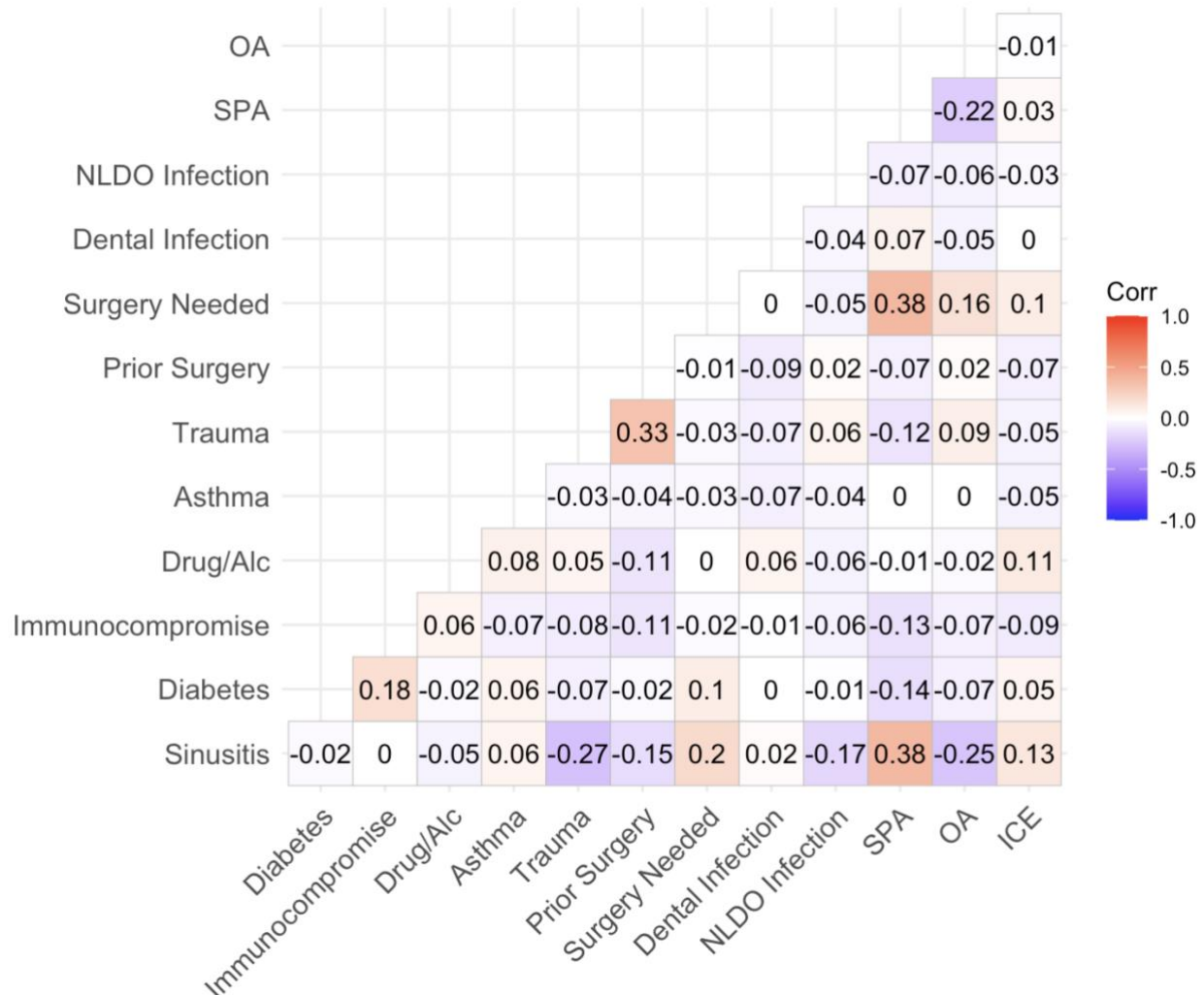


Figure 1. Pairwise correlation matrix of clinically relevant predictors and OC admissions. Heatmap showing Pearson correlation coefficients between clinical variables chosen a priori by PI. Colors represent the magnitude and direction of correlation (red = positive, blue = negative), with numerical values displayed within each cell.

Pearson correlation analysis was performed to assess associations between clinical predictors for modeling purposes. The analysis revealed weak to moderate correlations among several predictors. The strongest correlation was observed between Surgical Need and SPA/Sinusitis (ρ

= 0.38), with a moderately positive association. All other correlations observed were considered weak associations ($p < |0.35|$).

3.2 Associations Between Surgical Need and Predictors

Table 1: Factors Associated with Need for Surgery Among Adults Admitted for OC

Firth Logistic Regression Analysis

Predictor	N	OR	95% CI	%Δ in Odds	P-value
Age (per 1yr)	297	1.01	(0.99, 1.03)	1.1	0.194
Sex (Male vs Female)	297	1.22	(0.68, 2.22)	21.8	0.513
HL vs Blk non-HL	297	1.26	(0.34, 4.78)	25.9	0.729
White non-HL vs Blk non-HL)	297	1.37	(0.44, 4.53)	36.9	0.593
Sinusitis (Yes vs No)	297	1.47	(0.81, 2.72)	47.2	0.208
Diabetes (Yes vs No)	297	2.97	(1.46, 6.21)	196.9	0.002
Immuno (Yes vs No)	297	1.07	(0.51, 2.22)	7.0	0.855
Drugs/Alcohol (Yes vs No)	297	0.80	(0.35, 1.81)	-19.8	0.595
Asthma (Yes vs No)	297	0.73	(0.22, 2.33)	-26.8	0.598
Trauma (Yes vs No)	297	1.64	(0.75, 3.63)	64.4	0.213
Prior Surgery (Yes vs No)	297	0.89	(0.36, 2.15)	-11.2	0.792
Dental (Yes vs No)	297	0.90	(0.30, 2.69)	-9.8	0.852
NLDO (Yes vs No)	297	4.36	(0.51, 52.41)	335.5	0.172
SPA (Yes vs No)	297	38.44	(12.10, 194.21)	3,744.5	<0.001
OA (Yes vs No)	297	6.35	(2.96, 14.71)	534.5	<0.001
ICE (Yes vs No)	297	1.89	(0.43, 9.29)	88.6	0.399
Variables with very large odds ratios (e.g., SPA) reflect near-complete separation; effect sizes should be interpreted clinically rather than numerically.					

Association between clinical predictors and surgical need among adults admitted for orbital cellulitis.

Adjusted odds ratios (ORs) with 95% confidence intervals (CIs) were estimated using Firth logistic regression. Percent change in odds was calculated as $(OR - 1) \times 100$, where positive values indicate higher odds of surgical need and negative values indicate lower odds.

Patients with diabetes had nearly 3-fold higher odds of surgery (OR 2.97, 95% CI: 1.46 to 6.21, $p < 0.01$). Both SPA and OA demonstrated strong associations with surgical need. SPA showed a particularly large effect (OR 38.44, 95% CI 12.10–194.21, $p < 0.001$), consistent with near-complete separation and indicating that the vast majority of patients with SPA required surgery (65 out of 68). OA was also strongly associated with surgery (OR 6.35, 95% CI 2.96–14.71, $p < 0.001$).

3.3 Associations Between Length of Stay and Predictors

Table 2: Factors Associated with Length of Stay Among Adults Admitted for OC

Negative Binomial Regression Analysis

Predictor	N	RR	95% CI	%Δ in LOS	P-value
Age (per 1yr)	294	1.01	(1.00, 1.01)	0.5	0.035
Sex (Male vs Female)	294	0.87	(0.73, 1.03)	-12.8	0.119
HL vs Blk non-HL	294	1.58	(1.07, 2.34)	58.2	0.019
White non-HL vs Blk non-HL	294	1.13	(0.79, 1.61)	13.5	0.474
Sinusitis (Yes vs No)	294	1.16	(0.96, 1.40)	16.2	0.118
Diabetes (Yes vs No)	294	1.45	(1.16, 1.81)	44.9	0.001
Immuno (Yes vs No)	294	1.80	(1.46, 2.23)	80.0	<0.001
Drugs/Alcohol (Yes vs No)	294	1.24	(0.99, 1.56)	24.0	0.067
Asthma (Yes vs No)	294	0.87	(0.61, 1.24)	-13.2	0.424
Trauma (Yes vs No)	294	1.29	(1.01, 1.66)	29.5	0.040
Prior Surgery (Yes vs No)	294	0.73	(0.54, 0.98)	-27.2	0.035
Surgery During Admission (Yes vs No)	294	1.91	(1.58, 2.31)	90.7	<0.001
Dental (Yes vs No)	294	1.05	(0.77, 1.44)	4.8	0.765
NLDO (Yes vs No)	294	0.81	(0.35, 1.93)	-18.7	0.627
SPA (Yes vs No)	294	0.86	(0.68, 1.09)	-14.2	0.202
OA (Yes vs No)	294	0.99	(0.77, 1.26)	-1.4	0.908
ICE (Yes vs No)	294	2.09	(1.42, 3.14)	109.4	<0.001

Factors associated with hospital length of stay among adults admitted for orbital cellulitis. Rate ratios (RRs) and 95% confidence intervals (CIs) were estimated using negative binomial regression. RR represents the multiplicative change in expected length of stay associated with each predictor. Percent change in length of stay was calculated as $(RR - 1) \times 100$, where values greater than 1 indicate longer expected LOS and values less than 1 indicate shorter LOS.

Presence of an intracranial extension (ICE) was associated with a 109% longer expected LOS (RR 2.09; 95% CI: 1.42 to 3.14; $p < 0.001$). Other significant predictors of prolonged LOS included:

- Surgery during admission ~ 91% longer LOS
- Immunocompromised status (RR 1.80; 95% CI: 1.46 to 2.23; $p < 0.001$) ~80% longer LOS
- Diabetes (RR 1.45; 95% CI: 1.16 to 1.81, $p < 0.01$) ~45% longer LOS
- Trauma (RR 1.29; 95% CI: 1.01 to 1.66; $p < 0.05$) ~30% longer LOS

In contrast, prior surgery was associated with a 27% shorter LOS (RR 0.73; 95% CI: 0.54 to 0.98; $p < 0.05$).

3.4 Associations Between SPA and Predictors

Table 3: Factors Associated with SPA Among Adults Admitted for OC

Firth Logistic Regression Analysis

Predictor	N	OR	95% CI	%Δ in Odds	P-value
Age (per 1yr)	297	0.99	(0.97, 1.01)	-0.9	0.465
Sex (Male vs Female)	297	1.43	(0.60, 3.49)	42.5	0.425
HL vs Blk non-HL	297	1.92	(0.29, 12.43)	92.4	0.494
White non-HL vs Blk non-HL)	297	1.17	(0.21, 6.07)	17.1	0.854
Sinusitis (Yes vs No)	297	8.17	(2.78, 29.59)	716.7	<0.001
Diabetes (Yes vs No)	297	0.21	(0.06, 0.65)	-78.6	0.005
Immuno (Yes vs No)	297	0.24	(0.06, 0.77)	-75.7	0.015
Drugs/Alcohol (Yes vs No)	297	1.58	(0.47, 5.45)	58.5	0.455
Asthma (Yes vs No)	297	0.76	(0.13, 4.14)	-23.6	0.757
Trauma (Yes vs No)	297	0.50	(0.08, 2.56)	-49.6	0.414
Prior Surgery (Yes vs No)	297	1.98	(0.32, 11.12)	97.8	0.452
Surgery During Admission (Yes vs No)	297	37.41	(11.79, 187.95)	3,641.3	<0.001
Dental (Yes vs No)	297	1.11	(0.26, 5.29)	11.5	0.886
NLDO (Yes vs No)	297	0.83	(0.01, 15.00)	-16.9	0.912
OA (Yes vs No)	297	0.03	(0.00, 0.21)	-97.3	<0.001
ICE (Yes vs No)	297	0.43	(0.08, 2.11)	-57.1	0.295

Variables with very large odds ratios (e.g., surgery during admission) reflect near-complete separation; effect sizes should be interpreted clinically rather than numerically.

Association between clinical predictors and subperiosteal abscess (SPA) among adults admitted for orbital cellulitis. Adjusted odds ratios (ORs) with 95% confidence intervals (CIs) were estimated using Firth logistic regression. Percent change in odds was calculated as $(OR - 1) \times 100$, where positive values indicate higher odds of SPA and negative values indicate lower odds.

Among adults admitted for orbital cellulitis, sinusitis and surgery during admission were strongly associated with SPA.

Patients with sinusitis had over 8-fold higher odds of SPA (OR 8.17; 95% CI: 2.78 to 29.59; $p < 0.001$), while those undergoing surgery during admission had markedly higher odds (OR 37.41; 95% CI: 11.79 to 187.95; $p < 0.001$).

Conversely, diabetes (OR 0.21; 95% CI: 0.06 to 0.65, $p < 0.01$), immunocompromised status (OR 0.24; 95% CI: 0.06 to 0.77; $p < 0.05$), and OA (OR 0.03; 95% CI: 0.00 to 0.21; $p < 0.001$) were associated with lower odds of SPA.

3.5 Associations Between OA and Predictors

Table 4: Factors Associated with OA Among Adults Admitted for OC

Firth Logistic Regression Analysis

Predictor	N	OR	95% CI	%Δ in Odds	P-value
Age (per 1yr)	297	1.01	(0.98, 1.03)	0.5	0.620
Sex (Male vs Female)	297	0.43	(0.20, 0.90)	-57.3	0.025
HL vs Blk non-HL	297	1.79	(0.29, 19.70)	79.5	0.550
White non-HL vs Blk non-HL)	297	1.17	(0.24, 11.59)	16.9	0.865
Sinusitis (Yes vs No)	297	0.34	(0.14, 0.74)	-66.4	0.006
Diabetes (Yes vs No)	297	0.33	(0.10, 0.92)	-66.8	0.034
Immuno (Yes vs No)	297	0.55	(0.18, 1.45)	-45.5	0.233
Drugs/Alcohol (Yes vs No)	297	1.07	(0.34, 3.14)	7.2	0.902
Asthma (Yes vs No)	297	1.29	(0.27, 5.09)	28.5	0.735
Trauma (Yes vs No)	297	0.92	(0.34, 2.39)	-7.5	0.874
Prior Surgery (Yes vs No)	297	0.76	(0.22, 2.31)	-24.0	0.638
Surgery During Admission (Yes vs No)	297	5.73	(2.69, 13.17)	472.6	<0.001
Dental (Yes vs No)	297	0.86	(0.13, 3.87)	-14.5	0.850
NLDO (Yes vs No)	297	0.10	(0.00, 1.47)	-90.1	0.098
SPA (Yes vs No)	297	0.03	(0.00, 0.22)	-97.1	<0.001
ICE (Yes vs No)	297	0.37	(0.00, 3.86)	-63.2	0.465

Variables with very large odds ratios (e.g., surgery during admission) reflect near-complete separation; effect sizes should be interpreted clinically rather than numerically.

Association between clinical predictors and orbital abscess (OA) among adults admitted for orbital cellulitis. Adjusted odds ratios (ORs) with 95% confidence intervals (CIs) were estimated using Firth logistic regression. Percent change in odds was calculated as $(OR - 1) \times 100$, where positive values indicate higher odds of OA and negative values indicate lower odds

OA was associated with several clinical factors. Surgery during admission was strongly associated with OA (OR 5.73; 95% CI: 2.69 to 13.17; $p < 0.001$). In contrast, SPA was strongly inversely associated with OA (OR 0.03; 95% CI: 0.00 to 0.22; $p < 0.001$). Additionally, sinusitis (OR 0.34; $p < 0.01$) and diabetes (OR 0.33; $p < 0.05$) were associated with lower odds of OA. Males were also associated with reduced odds of OA (OR 0.43; $p < 0.05$).

3.4 Associations Between ICE and Predictors

Table 5: Factors Associated with ICE Among Adults Admitted for OC

Firth Logistic Regression Analysis

Predictor	N	OR	95% CI	%Δ in Odds	P-value
Age (per 1yr)	297	0.96	(0.92, 1.00)	-3.6	0.071
Sex (Male vs Female)	297	0.86	(0.22, 3.95)	-13.9	0.835
HL vs Blk non-HL	297	1.88	(0.05, 507.55)	87.8	0.754
White non-HL vs Blk non-HL)	297	5.46	(0.34, 1190.16)	445.6	0.281
Sinusitis (Yes vs No)	297	7.46	(1.08, 166.72)	645.9	0.041
Diabetes (Yes vs No)	297	6.29	(1.11, 40.06)	529.0	0.038
Immuno (Yes vs No)	297	0.10	(0.00, 0.98)	-89.9	0.047
Drugs/Alcohol (Yes vs No)	297	4.42	(1.00, 19.28)	342.1	0.051
Asthma (Yes vs No)	297	0.31	(0.00, 4.10)	-69.5	0.415
Trauma (Yes vs No)	297	1.90	(0.13, 32.08)	90.2	0.626
Prior Surgery (Yes vs No)	297	0.43	(0.00, 8.37)	-57.2	0.646
Surgery in Admission (Yes vs No)	297	1.50	(0.34, 7.27)	50.1	0.592
Dental (Yes vs No)	297	1.58	(0.14, 9.25)	57.5	0.664
NLDO (Yes vs No)	297	19.37	(0.08, 1923.73)	1,837.1	0.230
SPA (Yes vs No)	297	0.53	(0.09, 2.73)	-47.1	0.445
OA (Yes vs No)	297	0.47	(0.00, 5.92)	-52.8	0.607
Variables with very large odds ratios (e.g., surgery during admission) reflect near-complete separation; effect sizes should be interpreted clinically rather than numerically.					

Association between clinical predictors and intracranial extension (ICE) among adults admitted for orbital cellulitis. Adjusted odds ratios (ORs) with 95% confidence intervals (CIs) were estimated using Firth logistic regression. Percent change in odds was calculated as $(OR - 1) \times 100$, where positive values indicate higher odds of ICE and negative values indicate lower odds

Like SPA, patients with sinusitis had an approximately 8-fold higher odds of ICE (OR 7.46; 95% CI: 1.08 to 166.72; $p < 0.05$), but the estimate was not precise indicated by the wide confidence interval. Additionally, those with diabetes had 6-fold higher odds of ICE (OR 6.3; 95% CI: 1.11 to 40.06; $p < 0.05$). In contrast, those who were immunocompromised were associated with significantly lower odds of ICE (OR 0.1; 95% CI: 0.0 to 0.98; $p < 0.05$).

3.5 Sensitivity Analysis of Complication Variables

In sensitivity analyses excluding complication variables (SPA, OA, and ICE), several associations changed (**Supplementary Table 5 1**). Sinusitis became significantly associated with surgical need (OR 2.00; 95% CI: 1.20 to 3.34; $p < 0.01$), with an increased effect size compared to the fully adjusted model (**Table 1**). In contrast, the association between diabetes and surgical need was attenuated and no longer statistically significant. Immunocompromised status also demonstrated a change in direction, suggesting instability in estimates for this variable.

Associations in length of stay stayed relatively consistent with removal of complication variables, with the only exception being substance use becoming statistically significant (OR 1.34; 95% CI: 1.07 to 1.69; $p < 0.05$).

When evaluating the complication variables as outcomes, the association between sinusitis and each complication became stronger in magnitude. Sinusitis was more strongly positively associated with SPA and ICE, while its inverse associations with OA became more pronounced (**Supplementary Table 7 1**).

These findings suggest that the effects of certain predictors, particularly sinusitis, may be mediated through the development of complications.

Discussion

In this multi-center retrospective analysis of adults admitted for orbital cellulitis, several key clinical patterns were noted. Markers of disease severity, particularly subperiosteal abscess (SPA) and orbital abscess (OA), were the strongest predictors of surgical intervention. Diabetes was also associated with increased odds of surgery, suggesting that underlying metabolic disease may contribute to more severe infections. Other comorbidities and demographic factors were not significantly associated with surgical need after adjustment.

Length of stay (LOS) was driven by a different set of factors. Intracranial extension (ICE), immunocompromised status, diabetes, and surgical intervention during admission were all associated with prolonged hospital stay, likely reflecting greater disease severity and post operation recovery time. In contrast, SPA and OA were not significantly associated with LOS after adjustment, suggesting that while these complications influence surgical decision-making, they do not independently account for prolonged hospitalization.

Analyses of complication-specific outcomes revealed distinct clinical patterns. Sinusitis was strongly associated with SPA and ICE. In contrast, diabetes and immunocompromised status were associated with lower odds of SPA. OA demonstrated inverse associations with SPA, likely reflecting different disease progression rather than an independent process.

Sensitivity analyses further clarified these relationships. After removal of other complication variables, associations between sinusitis and outcomes increased in magnitude. Sinusitis showed a stronger positive association with SPA and ICE, while inverse association with OA became more pronounced. These findings suggest that sinusitis is most directly associated with SPA and ICE, whereas relationships with OA is more complicated may reflect different clinical pathways.

In contrast, the association between diabetes and surgical need was attenuated but remained elevated, suggesting both direct and indirect contributions to disease severity. Changes in direction observed for immunocompromised status likely reflect limited sample size and correlations with other severity-related variables.

Overall, these findings highlight the importance of considering disease pathway structure when interpreting multivariable models, as inclusion of downstream complications may attenuate or distort associations between baseline clinical factors and outcomes.

4.1 Limitations

This study has several important limitations. First, as a retrospective chart review, the analysis is subject to potential misclassification, missing data, and unmeasured confounding. Although complete-case analysis was used under a missing at random assumption, this assumption may not fully hold and could introduce bias.

Second, several predictors had low event frequencies, particularly for outcomes such as intracranial extension (ICE) and nasolacrimal duct obstruction (NLDO). This resulted in sparse data and potential instability in model estimates, as reflected by wide confidence intervals and evidence of near-complete separation for some variables. Although Firth penalized logistic regression was used to mitigate this issue, estimates for rare events should still be interpreted cautiously.

Third, several variables included in the models, particularly SPA, OA, and ICE, may lie along the same clinical disease pathway rather than representing independent predictors. As a result, models that include these complications simultaneously are typically subject to overadjustment or mediator adjustment, which could distort associations between upstream clinical factors and outcomes such as surgery or length of stay. Therefore, associations involving SPA, OA, and ICE should be interpreted as descriptive and hypothesis-generating rather than causal.

Fourth, the study population was derived from three hospital systems, which may limit generalizability to other settings with different patient populations.

Finally, the observational design precludes causal inference. Associations identified in this study should be interpreted as hypothesis-generating rather than definitive evidence of causal relationships.

4.2 Conclusions

In adults hospitalized with orbital cellulitis, associations with clinical outcomes showed distinct characteristics. Subperiosteal abscess and orbital abscess are the strongest predictors of surgical intervention, while intracranial extension, immunocompromised status, and diabetes are more strongly associated with prolonged length of stay.

Analyses of complication-specific outcomes suggest that SPA, OA, and ICE are related but distinct manifestations of disease severity, likely reflecting disease progression rather than independent risk factors.

These findings highlight the importance of differentiating between factors that drive surgical decision-making and those that influence overall disease burden. Inverse associations between SPA and OA/ICE should not be interpreted as evidence that SPA is protective but rather may reflect the sparse data or different stages of disease progression.

References

ChatGPT was used to check R code errors for some of the packages used and to grammar check (OpenAI, 2025).

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Supplementary Materials

****Supplementary Table 1: Characteristics of Orbital Cellulitis Adult Cohort****

Characteristic	N	Surgery Needed		
		Overall N = 330 ¹	No N = 137 ¹	Yes N = 193 ¹
Age	330			
Mean (SD)		49 (19)	49 (19)	50 (19)
Median [Q1, Q3]		49 [34, 64]	48 [32, 65]	51 [35, 64]
Min - Max		18 - 94	19 - 93	18 - 94
Sex	329			
Female		122 (37%)	54 (39%)	68 (35%)
Male		207 (63%)	83 (61%)	124 (65%)
NA		1	0	1
Race/Ethnicity	298			

Characteristic	N	Surgery Needed		
		Overall N = 330 ¹	No N = 137 ¹	Yes N = 193 ¹
<i>Black non-Hispanic</i>		19 (6.4%)	9 (7.3%)	10 (5.7%)
<i>Hispanic</i>		47 (16%)	18 (15%)	29 (17%)
<i>White non-Hispanic</i>		232 (78%)	96 (78%)	136 (78%)
<i>NA</i>		32	14	18
Sinusitis	330			
<i>No</i>		145 (44%)	76 (55%)	69 (36%)
<i>Yes</i>		185 (56%)	61 (45%)	124 (64%)
Diabetes	330			
<i>No</i>		272 (82%)	119 (87%)	153 (79%)
<i>Yes</i>		58 (18%)	18 (13%)	40 (21%)
Immunocompromised	330			
<i>No</i>		280 (85%)	115 (84%)	165 (85%)
<i>Yes</i>		50 (15%)	22 (16%)	28 (15%)
Drug/Alcohol	330			
<i>No</i>		280 (85%)	116 (85%)	164 (85%)
<i>Yes</i>		50 (15%)	21 (15%)	29 (15%)
Asthma	330			
<i>No</i>		311 (94%)	128 (93%)	183 (95%)
<i>Yes</i>		19 (5.8%)	9 (6.6%)	10 (5.2%)
Trauma	330			
<i>No</i>		283 (86%)	116 (85%)	167 (87%)
<i>Yes</i>		47 (14%)	21 (15%)	26 (13%)
Prior Surgery	330			
<i>No</i>		297 (90%)	123 (90%)	174 (90%)

Characteristic	N	Surgery Needed		
		Overall N = 330 ¹	No N = 137 ¹	Yes N = 193 ¹
Yes		33 (10%)	14 (10%)	19 (9.8%)
Dental	330			
No		308 (93%)	128 (93%)	180 (93%)
Yes		22 (6.7%)	9 (6.6%)	13 (6.7%)
NLDO	330			
No		323 (98%)	133 (97%)	190 (98%)
Yes		7 (2.1%)	4 (2.9%)	3 (1.6%)
SPA	330			
No		262 (79%)	134 (98%)	128 (66%)
Yes		68 (21%)	3 (2.2%)	65 (34%)
OA	330			
No		278 (84%)	125 (91%)	153 (79%)
Yes		52 (16%)	12 (8.8%)	40 (21%)
ICE	330			
No		315 (95%)	134 (98%)	181 (94%)
Yes		15 (4.5%)	3 (2.2%)	12 (6.2%)
Length of Stay	327			
Mean (SD)		7 (8)	5 (5)	9 (8)
Median [Q1, Q3]		5 [3, 9]	3 [2, 5]	7 [4, 11]
Min - Max		0 - 56	0 - 33	1 - 56
NA		3	1	2

¹n (%)

**** Supplementary Table 2: Characteristics of Orbital Cellulitis Adult Cohort****

Characteristic	N	SPA		
		Overall N = 330 ¹	No N = 262 ¹	Yes N = 68 ¹
Age	330			
<i>Mean (SD)</i>		49 (19)	51 (19)	42 (19)
<i>Median [Q1, Q3]</i>		49 [34, 64]	52 [35, 66]	38 [24, 57]
<i>Min - Max</i>		18 - 94	19 - 94	18 - 80
Sex	329			
<i>Female</i>		122 (37%)	103 (39%)	19 (28%)
<i>Male</i>		207 (63%)	159 (61%)	48 (72%)
<i>NA</i>		1	0	1
Race/Ethnicity	298			
<i>Black non-Hispanic</i>		19 (6.4%)	14 (5.9%)	5 (8.1%)
<i>Hispanic</i>		47 (16%)	36 (15%)	11 (18%)
<i>White non-Hispanic</i>		232 (78%)	186 (79%)	46 (74%)
<i>NA</i>		32	26	6
Sinusitis	330			
<i>No</i>		145 (44%)	140 (53%)	5 (7.4%)
<i>Yes</i>		185 (56%)	122 (47%)	63 (93%)
Diabetes	330			
<i>No</i>		272 (82%)	209 (80%)	63 (93%)
<i>Yes</i>		58 (18%)	53 (20%)	5 (7.4%)
Immunocompromised	330			
<i>No</i>		280 (85%)	216 (82%)	64 (94%)
<i>Yes</i>		50 (15%)	46 (18%)	4 (5.9%)
Drug/Alcohol	330			

Characteristic	N	SPA		
		Overall N = 330 ¹	No N = 262 ¹	Yes N = 68 ¹
No		280 (85%)	222 (85%)	58 (85%)
Yes		50 (15%)	40 (15%)	10 (15%)
Asthma	330			
No		311 (94%)	247 (94%)	64 (94%)
Yes		19 (5.8%)	15 (5.7%)	4 (5.9%)
Trauma	330			
No		283 (86%)	219 (84%)	64 (94%)
Yes		47 (14%)	43 (16%)	4 (5.9%)
Prior Surgery	330			
No		297 (90%)	233 (89%)	64 (94%)
Yes		33 (10%)	29 (11%)	4 (5.9%)
Dental	330			
No		308 (93%)	247 (94%)	61 (90%)
Yes		22 (6.7%)	15 (5.7%)	7 (10%)
NLDO	330			
No		323 (98%)	255 (97%)	68 (100%)
Yes		7 (2.1%)	7 (2.7%)	0 (0%)
Surgery During Admission	330			
No		137 (42%)	134 (51%)	3 (4.4%)
Yes		193 (58%)	128 (49%)	65 (96%)
OA	330			
No		278 (84%)	210 (80%)	68 (100%)
Yes		52 (16%)	52 (20%)	0 (0%)
ICE	330			
No		315 (95%)	251 (96%)	64 (94%)

Characteristic	N	SPA		
		Overall N = 330 ¹	No N = 262 ¹	Yes N = 68 ¹
Yes		15 (4.5%)	11 (4.2%)	4 (5.9%)
Length of Stay	327			
Mean (SD)		7 (8)	7 (8)	7 (6)
Median [Q1, Q3]		5 [3, 9]	5 [3, 9]	6 [4, 9]
Min - Max		0 - 56	0 - 56	2 - 40
NA		3	3	0

¹n (%)

****Supplementary Table 3: Characteristics of Orbital Cellulitis Adult Cohort****

Characteristic	N	SPA		
		Overall N = 330 ¹	No N = 278 ¹	Yes N = 52 ¹
Age	330			
Mean (SD)		49 (19)	49 (19)	54 (20)
Median [Q1, Q3]		49 [34, 64]	49 [33, 64]	55 [39, 68]
Min - Max		18 - 94	18 - 90	19 - 94
Sex	329			
Female		122 (37%)	94 (34%)	28 (54%)
Male		207 (63%)	183 (66%)	24 (46%)
NA		1	1	0
Race/Ethnicity	298			
Black non-Hispanic		19 (6.4%)	18 (7.1%)	1 (2.2%)

Characteristic	N	SPA		
		Overall N = 330 ¹	No N = 278 ¹	Yes N = 52 ¹
<i>Hispanic</i>		47 (16%)	39 (15%)	8 (17%)
<i>White non-Hispanic</i>		232 (78%)	195 (77%)	37 (80%)
<i>NA</i>		32	26	6
Sinusitis	330			
<i>No</i>		145 (44%)	107 (38%)	38 (73%)
<i>Yes</i>		185 (56%)	171 (62%)	14 (27%)
Diabetes	330			
<i>No</i>		272 (82%)	226 (81%)	46 (88%)
<i>Yes</i>		58 (18%)	52 (19%)	6 (12%)
Immunocompromised	330			
<i>No</i>		280 (85%)	233 (84%)	47 (90%)
<i>Yes</i>		50 (15%)	45 (16%)	5 (9.6%)
Drug/Alcohol	330			
<i>No</i>		280 (85%)	235 (85%)	45 (87%)
<i>Yes</i>		50 (15%)	43 (15%)	7 (13%)
Asthma	330			
<i>No</i>		311 (94%)	262 (94%)	49 (94%)
<i>Yes</i>		19 (5.8%)	16 (5.8%)	3 (5.8%)
Trauma	330			
<i>No</i>		283 (86%)	242 (87%)	41 (79%)
<i>Yes</i>		47 (14%)	36 (13%)	11 (21%)
Prior Surgery	330			
<i>No</i>		297 (90%)	251 (90%)	46 (88%)
<i>Yes</i>		33 (10%)	27 (9.7%)	6 (12%)

Characteristic	N	SPA		
		Overall N = 330 ¹	No N = 278 ¹	Yes N = 52 ¹
Dental	330			
No		308 (93%)	258 (93%)	50 (96%)
Yes		22 (6.7%)	20 (7.2%)	2 (3.8%)
NLDO	330			
No		323 (98%)	271 (97%)	52 (100%)
Yes		7 (2.1%)	7 (2.5%)	0 (0%)
Surgery During Admission	330			
No		137 (42%)	125 (45%)	12 (23%)
Yes		193 (58%)	153 (55%)	40 (77%)
SPA	330			
No		262 (79%)	210 (76%)	52 (100%)
Yes		68 (21%)	68 (24%)	0 (0%)
ICE	330			
No		315 (95%)	265 (95%)	50 (96%)
Yes		15 (4.5%)	13 (4.7%)	2 (3.8%)
Length of Stay	327			
Mean (SD)		7 (8)	7 (7)	8 (10)
Median [Q1, Q3]		5 [3, 9]	5 [3, 9]	5 [4, 7]
Min - Max		0 - 56	0 - 47	1 - 56
NA		3	3	0

¹n (%)

****Supplementary Table 4: Characteristics of Orbital Cellulitis Adult Cohort****

Characteristic	N	ICE		
		Overall N = 330 ¹	No N = 315 ¹	Yes N = 15 ¹
Age	330			

Characteristic	N	ICE		
		Overall N = 330 ¹	No N = 315 ¹	Yes N = 15 ¹
<i>Mean (SD)</i>		49 (19)	50 (19)	41 (16)
<i>Median [Q1, Q3]</i>		49 [34, 64]	50 [34, 65]	41 [24, 57]
<i>Min - Max</i>		18 - 94	18 - 94	20 - 62
Sex	329			
<i>Female</i>		122 (37%)	117 (37%)	5 (33%)
<i>Male</i>		207 (63%)	197 (63%)	10 (67%)
<i>NA</i>		1	1	0
Race/Ethnicity	298			
<i>Black non-Hispanic</i>		19 (6.4%)	19 (6.6%)	0 (0%)
<i>Hispanic</i>		47 (16%)	46 (16%)	1 (9.1%)
<i>White non-Hispanic</i>		232 (78%)	222 (77%)	10 (91%)
<i>NA</i>		32	28	4
Sinusitis	330			
<i>No</i>		145 (44%)	143 (45%)	2 (13%)
<i>Yes</i>		185 (56%)	172 (55%)	13 (87%)
Diabetes	330			
<i>No</i>		272 (82%)	261 (83%)	11 (73%)
<i>Yes</i>		58 (18%)	54 (17%)	4 (27%)
Immunocompromised	330			
<i>No</i>		280 (85%)	265 (84%)	15 (100%)
<i>Yes</i>		50 (15%)	50 (16%)	0 (0%)
Drug/Alcohol	330			
<i>No</i>		280 (85%)	270 (86%)	10 (67%)
<i>Yes</i>		50 (15%)	45 (14%)	5 (33%)

Characteristic	N	ICE		
		Overall N = 330 ¹	No N = 315 ¹	Yes N = 15 ¹
Asthma	330			
No		311 (94%)	296 (94%)	15 (100%)
Yes		19 (5.8%)	19 (6.0%)	0 (0%)
Trauma	330			
No		283 (86%)	269 (85%)	14 (93%)
Yes		47 (14%)	46 (15%)	1 (6.7%)
Prior Surgery	330			
No		297 (90%)	282 (90%)	15 (100%)
Yes		33 (10%)	33 (10%)	0 (0%)
Dental	330			
No		308 (93%)	294 (93%)	14 (93%)
Yes		22 (6.7%)	21 (6.7%)	1 (6.7%)
NLDO	330			
No		323 (98%)	308 (98%)	15 (100%)
Yes		7 (2.1%)	7 (2.2%)	0 (0%)
Surgery During Admission	330			
No		137 (42%)	134 (43%)	3 (20%)
Yes		193 (58%)	181 (57%)	12 (80%)
OA	330			
No		278 (84%)	265 (84%)	13 (87%)
Yes		52 (16%)	50 (16%)	2 (13%)
SPA	330			
No		262 (79%)	251 (80%)	11 (73%)
Yes		68 (21%)	64 (20%)	4 (27%)
Length of Stay	327			

Characteristic	N	ICE		
		Overall N = 330 ¹	No N = 315 ¹	Yes N = 15 ¹
Mean (SD)		7 (8)	7 (7)	16 (11)
Median [Q1, Q3]		5 [3, 9]	5 [3, 8]	13 [6, 24]
Min - Max		0 - 56	0 - 56	6 - 40
NA		3	3	0

¹n (%)

Supplementary Table 5 1

Need for Surgery					
term	OR	OR_ci	p_value	pct_chg	pct_ci
(Intercept)	0.55	(0.14, 2.24)	0.4066	-44.5	(-86.33, 124.28)
Age (per 1yr)	1.00	(0.99, 1.02)	0.5692	0.4	(-1.00, 1.86)
Sex (Male vs Female)	1.02	(0.61, 1.71)	0.9337	2.2	(-39.13, 71.11)
HL vs Blk non-HL	1.40	(0.47, 4.20)	0.5463	40.0	(-53.46, 319.80)
White non-HL vs Blk non-HL)	1.36	(0.52, 3.55)	0.5270	36.2	(-48.45, 255.01)
Sinusitis (Yes vs No)	2.00	(1.20, 3.34)	0.0073	99.7	(20.41, 233.92)
Diabetes (Yes vs No)	1.93	(1.00, 3.87)	0.0514	92.9	(-0.41, 287.19)
Immuno (Yes vs No)	0.72	(0.37, 1.38)	0.3193	-28.4	(-62.93, 38.41)
Drugs/Alcohol (Yes vs No)	0.85	(0.43, 1.69)	0.6351	-15.3	(-57.25, 68.91)
Asthma (Yes vs No)	0.63	(0.23, 1.71)	0.3609	-36.8	(-76.63, 70.68)
Trauma (Yes vs No)	1.31	(0.65, 2.69)	0.4558	30.9	(-35.42, 169.27)
Prior Surgery (Yes vs No)	0.81	(0.36, 1.83)	0.6099	-19.0	(-63.91, 83.09)
Dental (Yes vs No)	1.00	(0.40, 2.61)	0.9916	0.5	(-59.56, 161.14)
NLDO (Yes vs No)	1.79	(0.22, 20.53)	0.5807	78.9	(-77.75, 1952.95)

Supplementary Table 6

Length of Stay					
term	RR	RR_ci	p.value	pct_chg	pct_ci
Age (per 1yr)	1.00	(1.00, 1.01)	0.070	0.45	(-0.03, 0.94)
Sex (Male vs Female)	0.86	(0.72, 1.02)	0.084	-14.13	(-27.74, 1.91)

HL vs Blk non-HL	1.59	(1.06, 2.37)	0.021	58.81	(6.12, 136.73)
White non-HL vs Blk non-HL	1.19	(0.83, 1.71)	0.322	19.44	(-16.92, 70.71)
Sinusitis (Yes vs No)	1.18	(0.99, 1.41)	0.065	18.23	(-1.03, 41.18)
Diabetes (Yes vs No)	1.57	(1.27, 1.94)	0.000	56.55	(26.81, 93.88)
Immuno (Yes vs No)	1.72	(1.39, 2.13)	0.000	71.69	(39.15, 112.69)
Drugs/Alcohol (Yes vs No)	1.34	(1.07, 1.69)	0.013	34.22	(6.91, 69.27)
Asthma (Yes vs No)	0.82	(0.58, 1.18)	0.275	-17.80	(-42.19, 17.88)
Trauma (Yes vs No)	1.30	(1.01, 1.67)	0.042	29.70	(1.31, 66.60)
Prior Surgery (Yes vs No)	0.72	(0.53, 0.97)	0.031	-28.24	(-46.82, -2.88)
Surgery During Adm (Yes vs No)	1.81	(1.53, 2.14)	0.000	80.93	(52.83, 114.20)
Dental (Yes vs No)	1.12	(0.81, 1.54)	0.490	11.56	(-18.54, 54.28)
NLDO (Yes vs No)	0.84	(0.37, 2.03)	0.695	-15.57	(-63.31, 103.06)

Supplementary Table 7 1

SPA					
term	OR	OR_ci	p_value	pct_chg	pct_ci
(Intercept)	0.01	(0.00, 0.07)	0.000	-99.3	(-99.95, -92.51)
Age (per 1 yr)	0.99	(0.97, 1.01)	0.261	-1.2	(-3.36, 0.91)
Sex (Male vs Female)	1.42	(0.64, 3.20)	0.389	41.8	(-35.80, 220.34)
HL vs Blk non-HL	1.34	(0.23, 7.73)	0.742	34.0	(-76.88, 673.03)
White non-HL vs Blk non-HL)	0.84	(0.17, 3.93)	0.824	-16.1	(-82.63, 293.35)
Sinusitis (Yes vs No)	11.82	(4.42, 39.16)	0.000	1082.0	(342.37, 3815.86)
Diabetes (Yes vs No)	0.27	(0.08, 0.81)	0.018	-72.5	(-91.95, -19.33)
Immuno (Yes vs No)	0.34	(0.09, 1.04)	0.060	-66.3	(-90.78, 4.40)
Drugs/Alcohol (Yes vs No)	1.60	(0.51, 5.05)	0.418	60.2	(-49.36, 405.04)
Asthma (Yes vs No)	0.68	(0.13, 3.14)	0.630	-31.9	(-87.45, 214.38)
Trauma (Yes vs No)	0.39	(0.07, 1.73)	0.221	-61.2	(-92.91, 73.43)
Prior Surgery (Yes vs No)	2.06	(0.38, 10.01)	0.385	106.0	(-61.60, 901.29)
Surgery During Admi (Yes vs No)	29.14	(9.32, 145.20)	0.000	2814.4	(832.13, 14420.04)
Dental (Yes vs No)	1.08	(0.28, 4.29)	0.907	8.3	(-71.51, 329.45)
NLDO (Yes vs No)	1.92	(0.01, 31.84)	0.712	91.5	(-98.73, 3083.91)

Supplementary Table 8

OA					
term	OR	OR_ci	p_value	pct_chg	pct_ci

(Intercept)	0.19	(0.01, 1.67)	0.140	-80.7	(-98.60, 66.86)
Age (per 1 yr)	1.01	(0.99, 1.03)	0.339	1.0	(-1.04, 3.10)
Sex (Male vs Female)	0.41	(0.20, 0.85)	0.017	-58.7	(-80.18, -14.86)
HL vs Blk non-HL	1.77	(0.30, 18.92)	0.553	76.5	(-69.89, 1791.60)
White non-HL vs Blk non-HL)	1.12	(0.24, 10.88)	0.896	12.4	(-76.13, 987.68)
Sinusitis (Yes vs No)	0.17	(0.07, 0.37)	0.000	-83.1	(-92.63, -63.50)
Diabetes (Yes vs No)	0.36	(0.11, 0.97)	0.044	-64.2	(-88.89, -2.56)
Immuno (Yes vs No)	0.69	(0.23, 1.83)	0.474	-30.8	(-77.47, 83.41)
Drugs/Alcohol (Yes vs No)	0.96	(0.32, 2.55)	0.932	-4.4	(-67.78, 155.45)
Asthma (Yes vs No)	1.38	(0.30, 5.15)	0.658	37.5	(-70.04, 414.51)
Trauma (Yes vs No)	0.96	(0.36, 2.44)	0.938	-3.7	(-63.81, 143.77)
Prior Surgery (Yes vs No)	0.71	(0.21, 2.10)	0.549	-28.9	(-78.87, 110.14)
Surgery During Adm (Yes vs No)	4.15	(1.95, 9.60)	0.000	315.2	(94.60, 860.15)
Dental (Yes vs No)	0.74	(0.12, 2.96)	0.692	-26.2	(-87.63, 196.17)
NLDO (Yes vs No)	0.12	(0.00, 1.80)	0.134	-87.9	(-99.92, 79.92)

Supplementary Table 9

ICE					
term	OR	OR_ci	p_value	pct_chg	pct_ci
(Intercept)	0.01	(0.00, 0.64)	0.028	-99.2	(-100.00, -35.63)
Age (per 1 yr)	0.96	(0.92, 1.00)	0.050	-3.9	(-8.35, 0.01)
Sex (Male vs Female)	0.87	(0.22, 3.97)	0.843	-13.2	(-77.58, 296.61)
HL vs Blk non-HL	1.82	(0.05, 477.10)	0.769	81.6	(-94.87, 47609.87)
White non-HL vs Blk non-HL)	5.69	(0.35, 1208.12)	0.274	469.1	(-65.05, 120712.01)
Sinusitis (Yes vs No)	8.25	(1.27, 177.48)	0.024	725.1	(27.47, 17648.06)
Diabetes (Yes vs No)	9.82	(1.96, 56.85)	0.006	881.7	(96.49, 5585.35)
Immuno (Yes vs No)	0.11	(0.00, 1.05)	0.056	-89.2	(-99.92, 4.87)
Drugs/Alcohol (Yes vs No)	4.58	(1.02, 19.95)	0.047	357.8	(2.49, 1895.13)
Asthma (Yes vs No)	0.31	(0.00, 4.12)	0.423	-68.7	(-99.79, 312.47)
Trauma (Yes vs No)	1.90	(0.13, 22.73)	0.623	89.7	(-87.24, 2172.60)
Prior Surgery (Yes vs No)	0.45	(0.00, 9.54)	0.669	-55.3	(-99.85, 853.59)
Surgery During Adm (Yes vs No)	1.00	(0.28, 4.12)	0.999	0.1	(-71.81, 311.85)
Dental (Yes vs No)	1.69	(0.14, 10.16)	0.624	68.6	(-85.76, 916.03)
NLDO (Yes vs No)	32.25	(0.14, 3075.94)	0.160	3125.4	(-85.74, 307494.06)

Reproducibility Statement

The statistical code used in this analysis will be available on my GitHub directory at:

https://github.com/rysummers/Orbital-Cellulitis-Research/tree/main/adult_analysis

Raw data are not tracked in this repository and is available upon request.

Link to data on local drive: ' / Users/ryan_summers/Library/Mobile\ Documents/com\~apple\~CloudDocs/GitHub/Orbital-Cellulitis-Research/adult_analysis/DataRaw/OHSU\ and\ UC.xlsx'